Radial Basis Function (RBF) Kernel

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$$e^{-1/2||a-b||^2} = e^{-1/2}(a^Ta + b^Tb - 2a^Tb) = e^{-a^2/2}\, e^{-b^2/2}\, e^{a^Tb}$$

$$e^{a^Tb} = 1 + a^Tb/1! + (a^Tb)^2/2! + (a^Tb)^3/3! + (a^Tb)^4/4! + \ldots$$

$$\varphi(a)^T = e^{-a^2/2}\,[1,\ a_1/\sqrt{1}!,\ a_2/\sqrt{1}!,\ a_1^2/\sqrt{2}!,\ a_1a_2/\sqrt{2}!,\ a_2^2/\sqrt{2}!,\ \ldots]$$

$$\varphi(b) = e^{-b^2/2}\,[1,\ b_1/\sqrt{1}!,\ b_2/\sqrt{1}!,\ b_1^2/\sqrt{2}!,\ b_1b_2/\sqrt{2}!,\ b_2^2/\sqrt{2}!,\ \ldots]^T$$

Taylor series expansion of e^x

$$e^x = 1 + x/1! + x^2/2! + x^3/3! + x^4/4! + x^5/5! + \ldots$$

$$e^3 = 1 + 3/1! + 3^2/2! + 3^3/3! + 3^4/4! + 3^5/5! + \ldots$$

Approximations:

Approx. to 4 terms ⇒ 13

Approx. to 5 terms ⇒ 16.375

Approx. to 6 terms ⇒ 18.4

Approx. to 7 terms ⇒ 19.413

Approx. to 8 terms ⇒ 19.84

Approx. to 9 terms ⇒ 19.8420...

Actual Value ⇒ 20.086